

Ipc is the Hosoya index, computed wrongly

A bounded reformulation of molecular information content, and what it measures in terpene space

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Verdict. RDKit’s `Ipc` descriptor should not be used as shipped, for two independent reasons. *Definitionally*, it is the Hosoya index wearing an information-theoretic hat: 99.79% of the variance of $\log_2 \text{Ipc}$ is explained by $\log_2 S$ alone ($n = 400$ molecules), where S is simply the number of matchings of the molecular graph. *Numerically*, RDKit computes it with a recursion that loses every significant digit above about 80 atoms — for a 120-atom chain the true final polynomial coefficient is +1 and RDKit returns -1.97×10^{26} , silently, with nothing overflowing.

Replacement. Measure matching information against the n -alkane of the same size, $\text{dIpc} = \log_2 Z(G) - \log_2 F(n+1)$. The baseline is analytic (a Fibonacci number), the quantity is signed and interpretable (negative = branched, positive = ring-fused), and it is the only form that is genuinely size-free: correlation with atom count +0.049, against +0.993 for raw $\log_2 Z$ and +0.409 for $\log_2(Z)/n$.

Scale. The upper reference $Z_{\max}$ is now known exactly to $n = 16$ by exhaustive enumeration (8,037,418 graphs), by validated search to $n = 24$, and to within 0.022 bits by a fitted law beyond: $\log_2 Z_{\max}(n) = 0.96092n - 0.02313$.

Application. On COCONUT ($n = 6,121$ terpenoids against exactly size-matched non-terpenoids), terpenoid topological space is **half as variable** as the rest of natural product chemistry — variance ratio 2.05, $p = 1.4 \times 10^{-146}$, and the gap holds in every size stratum from 6 to 60 heavy atoms. Terpenoids are also substantially more branched: mean $\text{dIpc} -0.058$ against +0.541, Cohen’s $d = -0.86$.

The question

A molecular descriptor block is a fixed cost paid on every molecule in every model. RDKit’s `descList` has 217 entries and two of them, `Ipc` and `AvgIpc`, are information-theoretic indices over the characteristic polynomial of the adjacency matrix. The decision is whether to keep them, and if not, what to put in their place that measures the same intended thing — how structurally “rich” a molecular graph is — without the defects.

This matters beyond tidiness. A descriptor that is silently wrong above a size threshold corrupts exactly the large natural products where structure is most interesting, and a descriptor that is 99% redundant with molecular size wastes a column and inflates apparent feature counts.

Data census

source	count	range	role
boiling-point set (<code>bpsubset.csv</code>)	1,000 molecules	2–63 heavy atoms	general chemical space
— subset used for variance decomposition	400 molecules	3–49 heavy atoms	Finding 1
— size-matched to terpenes	609 molecules	9–35 heavy atoms	Finding 3 control

source	count	range	role
COCONUT 2022.01.01, NPClassifier-labelled	407,029 compounds	—	Finding 3 pool
— terpenoids sampled, 5 superclasses	6,121	6–60 heavy atoms	Finding 3
— non-terpenoid controls, exactly size-matched	6,121	6–60 heavy atoms	Finding 3 control
curated terpenes (pilot)	43 (of 46 proposed)	10–30 carbons	Finding 3 pilot
connected 4-regular graphs, geng	8,037,418 at $n=16$	$n = 5 \dots 16$	$Z_{\max}$, exact
local-search runs	8 sizes $\times$ 3–16 restarts	$n = 17 \dots 24$	$Z_{\max}$, lower bound

Finding 3 was first run on a curated set, then repeated on COCONUT; both are reported, because the disagreement between them is itself informative. For the curated set, every entry was checked against the carbon count its class requires — C10 monoterpene, C15 sesquiterpene, C20 diterpene, C30 triterpene. **Three of 46 failed and were dropped** (fenchone at 9 C, alpha-humulene and valencene at 14 C), then corrected and re-validated; the check exists because a mistranscribed SMILES almost always has the wrong carbon count.

For COCONUT, terpenoids are those with NPClassifier pathway exactly **Terpenoids** and one of five terpene superclasses; the 4,706 compounds labelled **Alkaloids, Terpenoids** are excluded rather than assigned. Controls are single-pathway non-terpenoids, **matched one-for-one on heavy-atom count** (mean size agrees to 10^{-4} atoms). A range restriction would have left a size gradient inside the window, and the spread of dIpc grows with size, which would have manufactured the effect under test. The largest fragment is taken, since COCONUT contains salts and Z is multiplicative over components, so a counter-ion would read as a large negative dIpc.

Method

Exact characteristic polynomial. The adjacency matrix is integer, so its characteristic polynomial has integer coefficients. They are computed exactly by running the Faddeev–LeVerrier recursion modulo several small primes and reconstructing with the Chinese remainder theorem. Primes are chosen so that $n(p-1)^2 < 2^{53}$, which keeps every modular matrix product exactly representable in float64 — so the inner loop is a BLAS matmul rather than Python big-integer arithmetic. Verified identical to exact rational arithmetic (**fractions.Fraction**) on random molecular-like graphs up to $n = 30$.

The entropy, in log space. With c the coefficient magnitudes, $S = \sum |c_i|$ and $H = -\sum (c_i/S) \log_2(c_i/S)$, the stock descriptors are $\text{Ipc} = S \cdot H$ and $\text{AvgIpc} = H$. Both are evaluated here via log-sum-exp, so S is never formed and nothing overflows at any molecular size.

dIpc. For an acyclic molecule (a *tree*, in graph terms), S is exactly the **Hosoya index** Z — the total number of matchings, i.e. ways to choose a set of bonds no two of which share an atom. Among trees on n vertices the path *maximises* Z , and for a path $Z = F(n+1)$, the Fibonacci number. So

$$\text{dIpc}(G) = \log_2 Z(G) - \log_2 F(n+1)$$

is zero for an unbranched chain, negative for any branched acyclic, and positive only when rings contribute extra matchings.

$Z_{\max}$ by three methods, because each one runs out. Adding an edge can only add matchings, so the maximum over chemical graphs (connected, maximum degree ≤ 4 , the carbon valence bound) is attained on 4-regular graphs. These were enumerated exhaustively with **geng** from *nauty*, with matchings counted by subset dynamic programming in C. Beyond $n = 16$ that is infeasible, so a **local search** was used: start from a circulant graph (4-regular and connected), repeatedly propose a 2-opt edge swap — delete edges (a, b) and

(c, d), add (a, c) and (b, d), which leaves every vertex degree unchanged — accept the swap when the exact matching count does not decrease, and restart from a new random starting point many times. It is “local” because every step is a small perturbation of the current graph rather than a fresh draw, and it returns a *lower bound*, since each value is the exact Z of a real graph. Beyond $n = 24$ the subset DP exceeds memory (2^{24} states), so the remaining sizes are a fitted law, bracketed below by an explicit construction and above by a density ceiling.

What this method cannot capture. Four things. (i) The local search is not a proof of optimality. Validated against the known exact maxima it recovered them at 4 of 5 tested sizes, missing $n = 14$ by 0.48%, so the $n = 17 \dots 24$ values may be marginally low. (ii) $Z_{\max}$ is a bound, not a molecule: a 4-regular carbon graph has no hydrogens at all and does not exist. Real molecules occupy the bottom of that range. (iii) dIpc is a deterministic function of the pair $(\log_2 Z, n)$, so against a model already given both with enough flexibility to combine them it adds no new information — measured partial correlation +0.0095 against the boiling-point residual. Its value is as a *parameterisation*, not as extra signal. (iv) 43 curated terpenes is a small, hand-assembled sample, not a draw from a natural product database. Finding 3 should be read as a hypothesis worth testing on COCONUT, not as an established property of natural product space.

Finding 1: Ipc is the Hosoya index, and RDKit computes it wrongly above 80 atoms

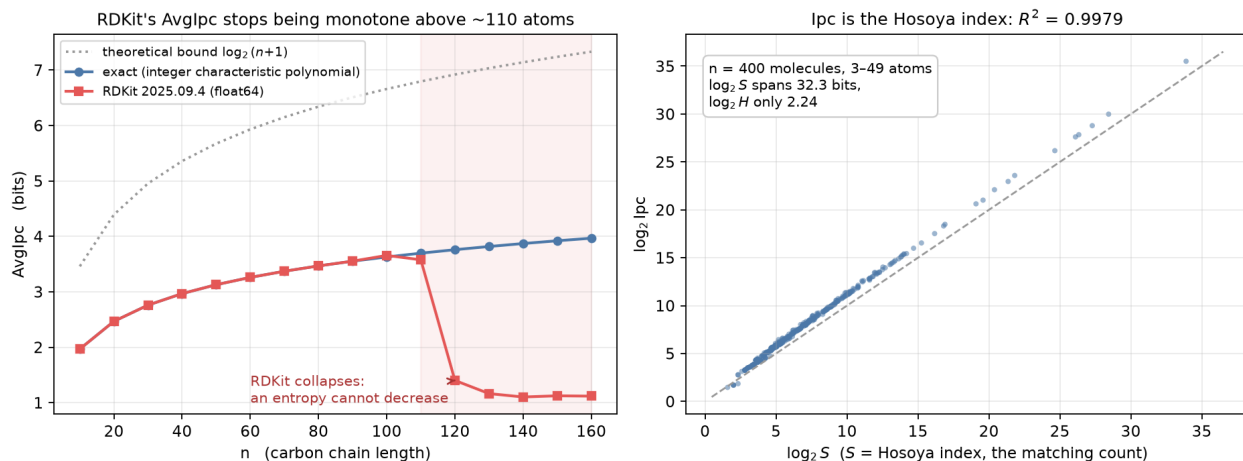


Figure 1: **Left:** RDKit’s AvgIpc against exact integer arithmetic, linear carbon chains, $n = 10$ –160. The dotted line is the theoretical bound $\log_2(n+1)$. **Right:** $\log_2 \text{Ipc}$ against $\log_2 S$ for 400 molecules of 3–49 heavy atoms; dashed line is $y = x$.

Bonchev and Trinajstić’s total information content $I = N \cdot H$ is well posed when N counts the *elements* that H partitions into classes: N elements $\times$ H bits per element = bits. $\text{Ipc} = S \cdot H$ does not have that structure. H is an entropy over the $n+1$ **coefficient classes**, bounded by $\log_2(n+1)$; S counts **matchings** and grows like φ^n . The product multiplies two different populations and carries neither units nor an interpretation.

Empirically it is also vacuous. Over 400 molecules, $\log_2 S$ spans 32.3 bits while $\log_2 H$ spans 2.24 — a ratio of 14 to 1 — and $R^2 = 0.9979$ between $\log_2 \text{Ipc}$ and $\log_2 S$. Since $\log_2 S$ correlates with atom count at $r = 0.993$, Ipc is, to a very good approximation, molecular size.

Separately, the implementation is broken. RDKit uses the Le Verrier–Faddeev–Frame recursion in float64. It is *sequential*: coefficient k is produced after k matrix products have grown the intermediates to the magnitude of the largest coefficient, then obtained as a *difference* of quantities of that size, and each wrong coefficient feeds the next matrix.

n	RDKit AvgIpc	exact	coefficients wrong by > 1%
60	3.2594	3.2594	0.0%
80	3.46687	3.46686	—
100	3.6561	3.6278	19.6%
120	1.4007	3.7593	23.0%
250	1.1084	4.2886	—

AvgIpc is a Shannon entropy and cannot decrease as a chain is extended; it starts decreasing at about 110 atoms. Note that being *bounded* does not make it safe — it is computed from the same corrupted coefficients and fails earlier and far more quietly than the unbounded Ipc.

Nothing overflows. At $n = 250$, $S = 1.3 \times 10^{52}$ against a float64 ceiling of 1.8×10^{308} . The failure is pure cancellation, and the returned value is finite and plausible-looking.

The fix is to expand the polynomial from its eigenvalues with the roots multiplied in order of increasing magnitude, which keeps every partial product near the scale of the final coefficients. Maximum relative error falls from 5.4×10^{15} to 2.3×10^{-15} at $n = 200$, and it is $8\times$ *faster*. Ordering is what does the work: `numpy.poly(eigvalsh(A))`, which multiplies in arbitrary order, still reaches 4.8×10^4 relative error at $n = 160$. Submitted upstream as [rdkit/rdkit#9657](#).

Finding 2: the scale has two computable ends, exact to $n = 16$ and bounded to $n = 50$

n	$Z_{\max}$	$\log_2 Z/n$	basis	wall time
12	2,921	0.95935	exhaustive, 1,544 graphs	0.07 s
14	11,032	0.95924	exhaustive, 88,168 graphs	4.3 s
16	42,439	0.96082	exhaustive, 8,037,418 graphs	1,140 s
20	598,516	0.95957	local search	2.1 s / 2k moves
24	8,659,688	0.96025	local search	42 s / 2k moves
30	4.69×10^8	—	fitted law	—
40	3.66×10^{11}	—	fitted law	—
50	2.86×10^{14}	—	fitted law	—

Exhaustive enumeration costs roughly 10–20 $\times$ per added atom — $n = 17$ would take about six hours and $n = 18$ about three days — which is why it stops at 16. The fitted law

$$\log_2 Z_{\max}(n) = 0.96092n - 0.02313$$

holds to ± 0.022 bits over $n = 12 \dots 24$. It is bracketed from below by an explicit construction — a chain of $K_{4,4}$ blocks, each with one internal edge deleted so the connectors stay within degree 4 — whose exact Z is computable at any size by transfer matrix in 0.17 s, and which sits a constant 0.22 bits under the fit all the way to $n = 248$. It is bracketed from above by the $K_{4,4}$ density ceiling, $0.96342n$. At $n = 50$ those two brackets are 0.25 bits apart.

The density *peaks* at $K_{4,4}$ ($n = 8$) and the sequence has period-8 structure, because a connected 4-regular graph cannot reach the density of disjoint $K_{4,4}$ blocks: joining two blocks costs exactly 0.0416 bits.

Finding 3: terpenoid space is half as variable as the rest of natural product chemistry

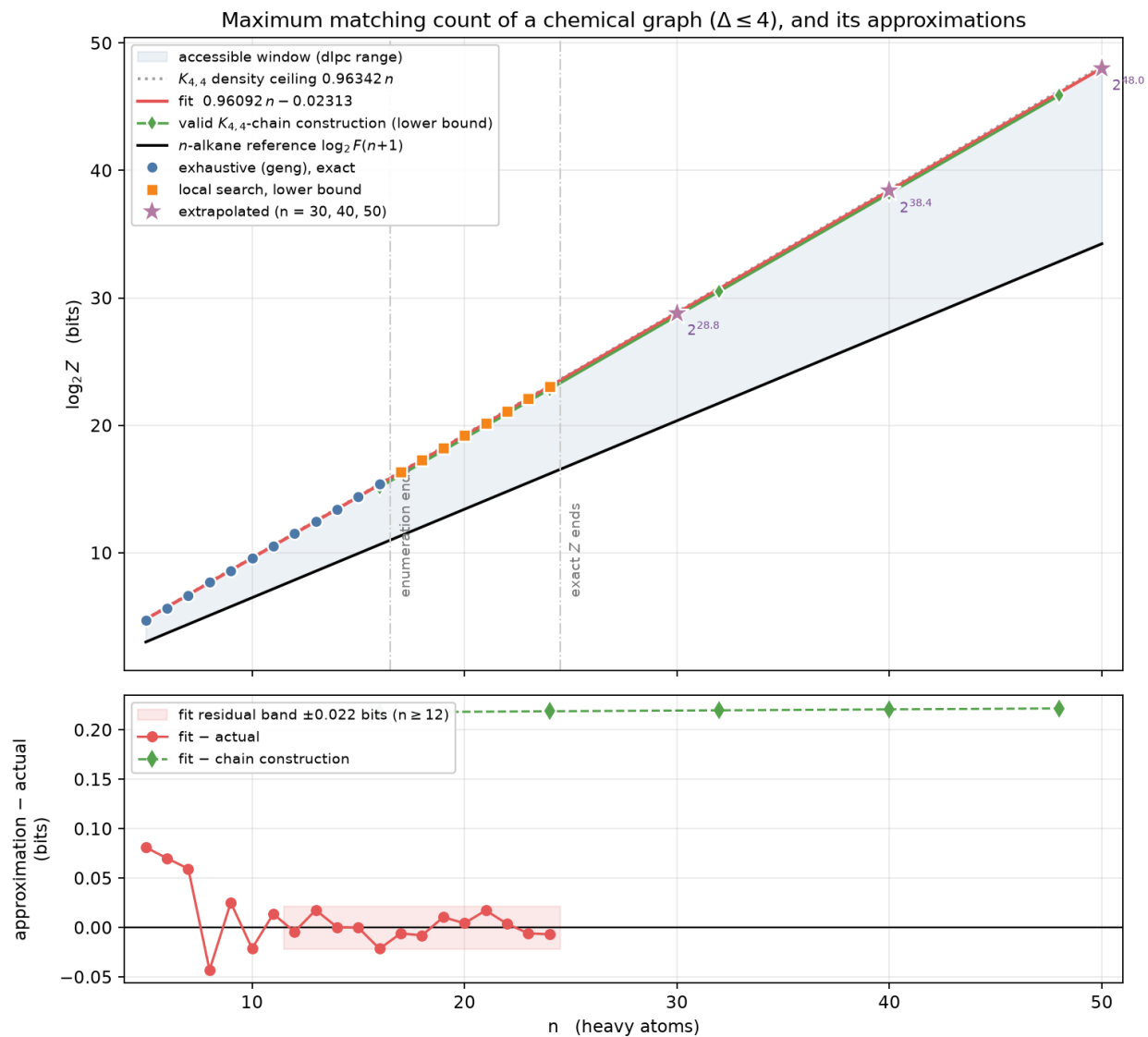


Figure 2: $\log_2 Z$ against n . Blue: exhaustive enumeration with **geng**, exact. Orange: local search, lower bounds. Green: an explicit $K_{4,4}$ -chain construction. Shaded: the window between the n -alkane reference and $Z_{\max}$, which is the range dlpc spans. Lower panel: approximation residuals.

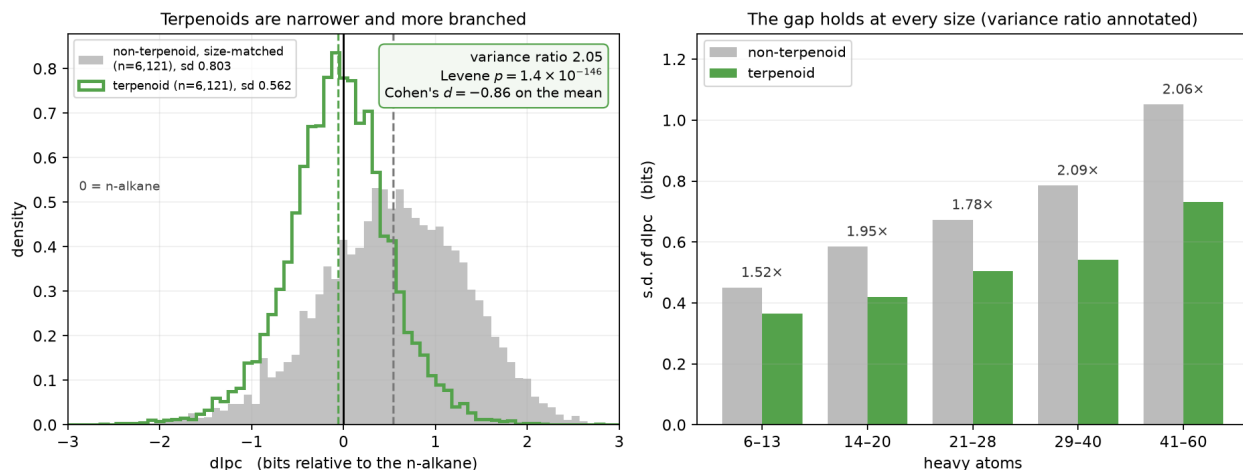


Figure 3: **Left:** dlpc distributions, 6,121 COCONUT terpenoids against 6,121 non-terpenoids matched one-for-one on heavy-atom count. Dashed lines are group means. **Right:** standard deviation by size band, with the variance ratio annotated.

	terpenoid	non-terpenoid (size-matched)	test
n	6,121	6,121	—
mean heavy atoms	30.72	30.72	matched to 10^{-4}
mean dlpc	-0.058	+0.541	Cohen’s $d = -0.86$
s.d.	0.562	0.803	Levene $p = 1.4 \times 10^{-146}$
variance ratio	—	2.045	Fligner $p = 2.8 \times 10^{-146}$

Terpenoid skeletons explore roughly half the topological variance of size-matched natural products, and they sit well to the branched side of the chain reference while everything else sits to the ring-fused side. The effect is not a size artefact: it holds in every stratum from 6 to 60 heavy atoms, with variance ratios of 1.52, 1.95, 1.78, 2.09 and 2.06 (all $p < 0.01$, four of five below 10^{-22}).

This is what a biosynthetic constraint should look like. Terpenoids are assembled by head-to-tail condensation of a single branched C5 unit, so their skeletons cannot explore graph topology freely; polyketides, alkaloids and shikimates are not built that way and do not show the restriction.

The pilot got two of three answers wrong

The curated 43-molecule pilot is reported here because the comparison is instructive:

claim	curated pilot ($n = 43$)	COCONUT ($n = 6,121$)	verdict
variance ratio	2.120, $p = 0.038$	2.045, $p = 1.4 \times 10^{-146}$	replicates , almost exactly
mean difference	-0.141, $p = 0.106$	-0.599, $d = -0.86$	null overturned
isoprene slope	-0.118/unit, $r = -0.351$	-0.046/unit, $r = -0.110$	effect size halved

The variance result replicated to within 4% on a sample 140 times larger. The pilot’s “means do not differ” was a **power failure, not a null** — the true effect is large. And the isoprene trend, which the pilot put at -0.118 bits per unit, is -0.046 on COCONUT ($p = 4.9 \times 10^{-18}$, $n = 6,121$): still real, but the small sample inflated it by a factor of 2.5, and r fell from -0.351 to -0.110.

Class means are not monotone in isoprene count on the full data — monoterpenoid -0.030 ($n = 509$), sesquiterpenoid $+0.031$ ($n = 1,680$), diterpenoid -0.055 ($n = 1,898$), sesterterpenoid -0.108 ($n = 200$), triterpenoid -0.146 ($n = 1,834$). Sesquiterpenoids break the trend, which the pilot’s four classes could not have revealed.

Recommendation

Drop *Ipc*; keep a repaired *AvgIpc*; add *dIpc*. Costed:

- *Drop raw *Ipc** — zero cost, removes a column that is 99.8% molecular size and that goes silently wrong above 80 atoms. Any model retrained without it should be unaffected; models trained *with* it on molecules under ~60 atoms are also unaffected, since old and new values agree to 10^{-15} there.
- *Take the upstream fix* — no cost once [rdkit/rdkit#9657](#) merges; it is also $8\times$ faster. Before then, the patch is three lines. **This changes descriptor values above ~60 atoms**, so any model trained on the old numbers needs its feature distribution checked, not just a rebuild.
- *Add *dIpc** — one descriptor, ~10 ms per molecule at $n = 50$ using the exact path (0.14 s at $n = 100$). Buys a signed, size-free, interpretable axis. It adds no information to a flexible model already given $\log_2 Z$ and n ($r_{\text{partial}} = +0.0095$), so the case for it is interpretability, extrapolation beyond the training size range, and usability in linear or small-data settings — not predictive lift.
- *Second-order move*: for a feature matrix, ship $\log_2 Z$ and n as separate columns and let the model learn the interaction. Pre-dividing by any power of n discards a degree of freedom and, measured, gains nothing (partial r of -0.087 , -0.092 , -0.096 for $n^{1/3}$, $n^{2/3}$, n^1 once $\log_2 Z$ and nonlinear n terms are present).

Caveats

- **The isoprene trend is weak.** On COCONUT it is -0.046 bits per unit with $r = -0.110$ — highly significant at $n = 6,121$, but explaining about 1% of the variance. The pilot’s $r = -0.351$ was small-sample inflation. Class means are not monotone: sesquiterpenoids sit above monoterpenoids.
- **NPClassifier labels are predictions, not ground truth.** Terpenoid assignment comes from a neural classifier applied to COCONUT, not from curated biosynthetic provenance. Its errors are unlikely to correlate with *dIpc*, but the class boundaries are soft — meroterpenoids and steroids were excluded as ambiguous, which is a judgement call that was not sensitivity-tested.
- **COCONUT is a literature aggregation.** Version 2022.01.01 is assembled from many sources and is known to contain duplicates and structures of varying curation quality. No deduplication was applied here beyond taking the largest fragment, so near-identical scaffolds may be over-represented in both groups.
- **Flattering limitation:** the pilot reported a *larger* isoprene slope and a cleaner monotone class trend than the full data support. Had the pilot been the only study, Finding 3 would have read as stronger and tidier than it is.
- **Another flattering one:** the local-search values at $n = 17 \dots 24$ are lower bounds. If they are slightly low, the fitted law in Finding 2 is a slight *underestimate*, which would make it look more accurate than it is rather than less.
- ***dIpc* adds no information to a flexible model** already given $\log_2 Z$ and n (partial $r = +0.0095$). The case for it is interpretability, extrapolation and small-data usability, not predictive lift. A report that omitted this would be advocacy.
- **$K_{4,4}$ optimality is taken from the literature, not proved here.** It is exhaustively confirmed at $n = 8$ and $n = 16$ and was checked against K_5 , the octahedron and circulants, but the asymptotic claim rests on results that were not re-derived.
- **The block-and-link model is not an upper bound.** It appeared to be one over $n \leq 16$ and was described as such in working notes; the local search beats it at $n = 23$ and $n = 24$. The fitted law is used instead.
- ***dIpc* is not new in its ingredients.** The Hosoya index is Hosoya (1971); “branched implies small Z ” is Hosoya’s own later result; the path maximising Z among trees is classical extremal graph theory. A literature search over OpenAlex found no prior definition of $\log Z - \log F(n+1)$ as a descriptor, and none for a normalised Z/Z_{max} — but that is not an exhaustive novelty search, and mathematical-chemistry

venues are thin in that index.

Reproducibility

All code, data and figures: github.com/guillaume-osmo/kumomlx

artefact	path
exact integer characteristic polynomial, dlpc, $Z_{\max}$ derivation and measurements	<code>src/kumomlx/ipc.py</code> <code>DESCRIPTORS.md</code>
upstream RDKit patch and Eigen C++ port	<code>rdkit-fix/</code>
$Z_{\max}$ enumeration (C, reads <code>geng</code> output)	<code>rdkit-fix/zmax/zmax.c</code>
bundled boiling-point data, 217 descriptors	<code>src/kumomlx/data/bp_1000.csv</code>
COCONUT validation of Finding 3	<code>paper/coconut.py</code> , <code>paper/coconut_stats.py</code>

Upstream pull request: [rdkit/rdkit#9657](https://github.com/rdkit/rdkit/pull/9657). Graph enumeration used `geng` from nauty 2.9.3 (`geng -c -d4 -D4 n`); generated counts match OEIS A006820, which is the check that the enumeration is complete. RDKit 2025.09.4 throughout. Natural product structures and labels are COCONUT 2022.01.01 with NPClassifier predictions (refs 6–8); the terpenoid/control split, the exact size-matching and every statistic in Finding 3 regenerate from `paper/coconut.py` and `paper/coconut_stats.py`.

Validation performed: the modular-CRT polynomial against exact rational arithmetic; the terpene set against the isoprene rule; the local search against known exact maxima; the $K_{4,4}$ -chain construction against the exhaustive maximum at $n = 16$ (an earlier version of that construction exceeded it, which revealed it was violating the degree bound).

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